Binomial Coefficients as a Calculus of Finite Choices

N-20221210

The binomial coefficient $\binom{n}{k}$ counts the k-element subsets of an n-element set. That familiar count becomes much more powerful when it is treated as a change of coordinates: a function of an integer can be expanded in the basis $\binom{x}{k}$, just as a smooth function is expanded in powers of x. The inverse operation is a finite difference, and the same subset counting leads to inclusion–exclusion, probability, and labelled combinatorial structures.[1]

§1.1 The binomial expansion is a decomposition into choices

Expanding $(a+b)^n$ means choosing either a or b from each of n factors. If exactly k factors contribute a, the resulting monomial is $a^k b^{n-k}$, and there are $\binom{n}{k}$ choices. Therefore

$$\boxed{(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.}$$

Putting $a = b = 1$ gives $\sum_k \binom{n}{k} = 2^n$: every subset is counted once. Putting $a = 1, b = -1$ gives, for $n \geq 1$,

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0.$$

Thus the even-sized and odd-sized subsets are equally numerous. There is also a direct bijection: choose a fixed element and toggle whether it belongs to the subset, thereby reversing its cardinality's parity.

The triangle inequality yields the useful bound

$$|a+b|^n \leq (|a| + |b|)^n = \sum_{k=0}^{n} \binom{n}{k} |a|^k |b|^{n-k}.$$

The validity of the inequality does not depend on formally moving absolute values through an equality; it follows from the ordinary triangle inequality applied to $a + b$.

§1.2 Inclusion–exclusion is a binomial cancellation

For finitely many sets $A_1, \ldots, A_n$ inside a finite universe U,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

To prove it, fix $u \in U$ belonging to exactly r of these sets. It is counted on the right with total weight

$$\sum_{k=1}^{r} (-1)^{k+1} \binom{r}{k} = 1 - (1-1)^r.$$

[1] The binomial-basis and combinatorial generating-function perspective follows Stanley, *Enumerative Combinatorics*, Vol. 1, Chapters 1 and 3, and Flajolet–Sedgewick, *Analytic Combinatorics*, Chapter II: https://algo.inria.fr/flajolet/Publications/book.pdf.

The weight is 1 if $r \geq 1$ and 0 if $r = 0$, exactly the indicator of membership in the union.

This computation is the elementary core of Möbius inversion on the Boolean lattice of subsets: alternate signs cancel all the overcounted choices.

§1.3 Binomial inversion recovers hidden coefficients

Suppose two sequences are related by

$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k.$$

Then the inverse transform is

$$\boxed{a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k.}$$

To verify it, substitute the first relation into the second and reverse the finite sums:

$$\begin{aligned}\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k &= \sum_{j=0}^{n} a_j \sum_{k=j}^{n} (-1)^{n-k} \binom{n}{k} \binom{k}{j} \\ &= \sum_{j=0}^{n} a_j \binom{n}{j} \sum_{r=0}^{n-j} (-1)^{n-j-r} \binom{n-j}{r}.\end{aligned}$$

The last inner sum is $(1-1)^{n-j}$. Only the term $j = n$ survives, giving a_n.

For instance, if $b_n = 2^n$, the inverse formula yields

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} 2^k = (2-1)^n = 1.$$

The direct transform of the constant sequence $a_k = 1$ is indeed 2^n.

§1.4 Finite differences take the place of derivatives

For a function $f : \mathbb{Z}_{\geq 0} \to \mathbb{R}$, define its forward difference

$$\Delta f(n) = f(n+1) - f(n).$$

Iterating produces

$$\boxed{\Delta^k f(0) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} f(j).}$$

The coefficients are the same as in binomial inversion. The analogy with differentiation becomes exact on the binomial polynomials

$$\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}.$$

Pascal's identity says

$$\Delta \binom{x}{k} = \binom{x}{k-1},$$

where $\binom{x}{0} = 1$. Thus the binomial basis is adapted to discrete differentiation, much as $x^k/k!$ is adapted to ordinary differentiation.

If $f(x)$ is a polynomial of degree at most d, its Newton expansion is

$$\boxed{f(x) = \sum_{k=0}^{d} \Delta^k f(0) \binom{x}{k}.}$$

Indeed, successive finite differences of this expression at 0 agree with those of f; two polynomials of degree at most d with the same first $d+1$ Newton coefficients are identical.

Take $f(x) = x^3$. Directly,

$$\Delta f(0) = 1, \quad \Delta^2 f(0) = 6, \quad \Delta^3 f(0) = 6.$$

Hence

$$\boxed{x^3 = \binom{x}{1} + 6\binom{x}{2} + 6\binom{x}{3}.}$$

This identity is a discrete analogue of Taylor's theorem. It expresses the cubic in basis functions that respond to Δ by lowering their index.

§1.5 Summation is the inverse of taking differences

Pascal's identity also gives

$$\binom{k+1}{r+1} - \binom{k}{r+1} = \binom{k}{r}.$$

Summing from $k = 0$ to $n-1$ telescopes:

$$\boxed{\sum_{k=0}^{n-1} \binom{k}{r} = \binom{n}{r+1}.}$$

This is the hockey-stick identity. In discrete calculus, the binomial polynomial $\binom{x}{r+1}$ plays the role of an antiderivative of $\binom{x}{r}$.

Combining this with the Newton expansion immediately evaluates sums of polynomial sequences. Using the cubic identity above,

$$\sum_{k=0}^{n-1} k^3 = \binom{n}{2} + 6\binom{n}{3} + 6\binom{n}{4} = \left(\frac{n(n-1)}{2}\right)^2.$$

Thus familiar power-sum formulas are instances of a systematic discrete integration method.

§1.6 The same coefficients form a probability distribution

Let X be the number of selected elements when each of n labels is independently included with probability p. Then

$$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The generating polynomial is

$$\mathbb{E}[z^X] = (1 - p + pz)^n.$$

Differentiating at $z = 1$ gives the factorial moments:

$$\mathbb{E}[X] = np, \qquad \mathbb{E}[X(X-1)] = n(n-1)p^2.$$

Since $X^2 = X(X-1) + X$, we obtain

$$\boxed{\mathrm{Var}(X) = np(1-p).}$$

For $p = 1/2$, the central binomial coefficients control the most likely subset sizes. Chebyshev's inequality already gives a quantitative concentration estimate:

$$\Pr\left(\left|\frac{X}{n} - \frac{1}{2}\right| \geq \varepsilon\right) \leq \frac{1}{4n\varepsilon^2}.$$

The binomial theorem therefore simultaneously normalizes a probability distribution and gives access to its fluctuations.

§1.7 From choosing subsets to assembling structures

Subset counting asks us to label each element in one of two ways: selected or not selected. Other labelled structures can be assembled from components, with a generating function recording the number of ways to make each choice.

A permutation decomposes uniquely into disjoint cycles. The number of cycles of length k on a fixed k-element label set is $(k-1)!$, so the exponential generating function of one cycle is

$$C(z) = \sum_{k\geq 1} \frac{(k-1)!}{k!} z^k = \sum_{k\geq 1} \frac{z^k}{k} = -\log(1-z).$$

A permutation is an *unordered set of cycles*. The exponential formula for labelled structures therefore gives

$$P(z) = \exp(C(z)) = \frac{1}{1-z}.$$

If p_n denotes the number of permutations of n labels, the coefficient convention is $P(z) = \sum_{n\geq 0} p_n z^n/n!$, so $p_n = n!$, as expected.

The exponential formula can be checked directly in this case: expanding $\exp(C(z))$ counts unordered collections of cycles, and the division by $r!$ in the exponential removes the ordering of r components. The division by $k!$ in the generating function records the relabelling of an individual size-k component. The algebra captures precisely the combinatorial symmetries involved.

Binomial coefficients begin as counts of subsets, but their usefulness comes from several compatible structures. Alternating signs implement inversion; finite differences provide discrete derivatives; generating functions record independent choices; and exponential generating functions account for the relabelling of assembled components. The same small counting identity has grown into a calculus for finite structures.